

GRACE WOU

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EDUCATION

University of Michigan, Bachelor of Science in Engineering

Aug. 2024 – May 2028

Mechanical Engineering Major, Robotics Concentration

Ann Arbor, MI

EXPERIENCE

MedLaunch

Sep. 2024 – Present

Internal Communications Chair

Ann Arbor, MI

- Engaging in year-long biodesign projects developing assistive technologies for community partners with disabilities.
- Developed a custom sports vest for an amputee soccer goalkeeper by creating sewing patterns that optimize mobility and structure, researching fabric materials, and rapid prototyping the vest through sewing and functional testing.
- Designed and built an adaptive wheelchair device by implementing quick-attach wheelchair mechanisms, evaluating surface concavities for optimal hitting surface and power, and testing solutions with athletes in game-like scenarios.

Boston Scientific

May 2026 – Aug. 2026

Process Development Intern

Arden Hills, MN

- Designed and developed a custom PCB fixture from concept through CAD, rapid prototyping, and 3D printing, increasing usable board capacity by 57% while maintaining support for capacitor terminal pins and improving operator usability.
- Led a process improvement DMAIC investigation targeting capacitor electrolyte weight-gain variability historically responsible for 7% rework, utilizing Ishikawa diagrams, statistical analysis, microscopy inspection, and controlled experimental testing to identify critical process drivers.

Harbinger Motors

May 2025 – Aug. 2025

Quality Engineering Intern

Orange County, CA

- Reduced chassis defects by 48% by applying quality engineering methodologies, including 4 Diamonds and Kaizen.
- Conducted root cause analysis of a wheel bearing NCR by utilizing CAD models and work instructions, transitioning from force-based to distance-based pressing to improve consistency and capability of the wheel bearing press.
- Designed and fabricated precision gauges for wheel bearing measurements with CAD and 3D printing, reducing frequency of inspection processes by 90%.
- Operated 3D printing equipment to fabricate 10+ prototype components for design iterations.

PROJECTS

Plants vs. Zombies

Sep. 2026 – Present

Design and Manufacturing II

- Designing a powered target-return mechanism using light sensors to detect and redirect targets along four rails.
- Designed and analyzed a 4-bar linkage and transmission using SolidWorks and ADAMS, evaluating kinematic and dynamic performance such as torque and inertia to determine an optimal transmission ratio.

Guitar Pedal

Sep. 2026 – Present

Personal Project

- Building a functional guitar pedal circuit, applying principles of circuit design, soldering, and electronic troubleshooting.
- Designing and fabricating a custom pedal enclosure with CAD and 3D printing, incorporating component placement, mounting features, and accessibility for assembly.

Robotic Machine Player

Sep. 2025 – Dec. 2025

Design and Manufacturing I

- Engineered a rack-and-pinion linear motion system and custom forklift mechanism in SolidWorks, producing detailed CAD models, assemblies, and fabrication drawings.
- Manufactured 10+ parts using a manual mill and lathe while achieving tolerances of 0.001".
- Performed mechanical assembly, electrical wiring, and soldering to integrate hardware subsystems into functioning robot.

TECHNICAL SKILLS

Languages: Python, Java, C++, MATLAB

CAD/Fabrication: SolidWorks, CATIA, 3D Printing, Mill, Lathe, Waterjet, Soldering

Tools: NetSuite, Jira, Microsoft Excel, Minitab, MSC Adams